

Class Activity

1) Hospital Patient Admission Analysis

Create a Dataset:

```
# Create Dataset
```

```
hospital <- data.frame(  
  Department = c("Cardiology",  
                 "Neurology",  
                 "Pediatrics",  
                 "Orthopedics",  
                 "Emergency"),  
  Admissions = c(120, 90, 150, 80, 200)  
)
```

```
print(hospital)
```

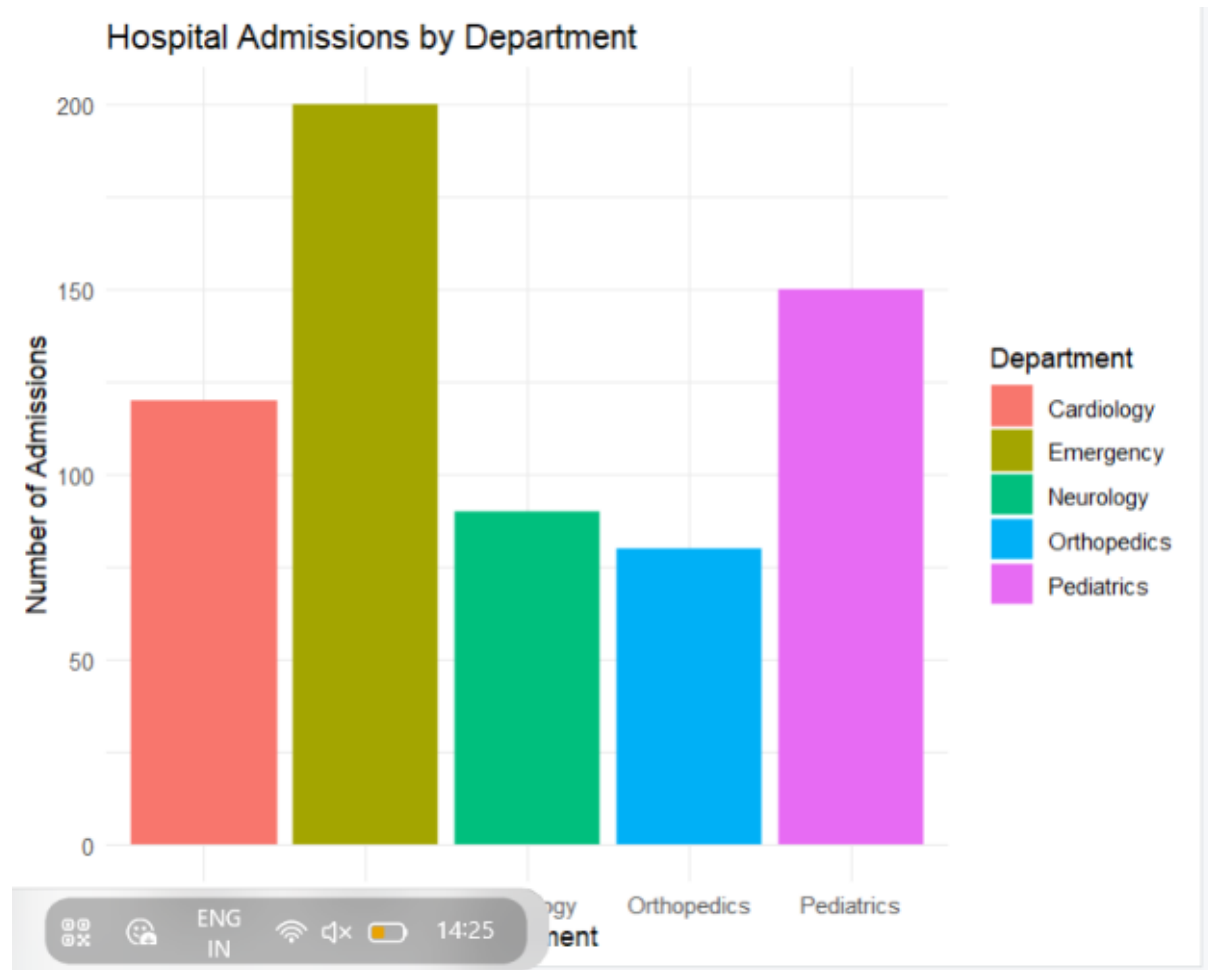
Bar chart :

```
library(ggplot2)
```

```
ggplot(hospital,  
  aes(x = Department,  
      y = Admissions,  
      fill = Department)) +  
  geom_col() +  
  labs(  
    title = "Hospital Admissions by Department",  
    x = "Department",  
    y = "Number of Admissions"  
  ) +
```

```
theme_minimal()
```

output:



Stacked bar chart:

```
library(ggplot2)
```

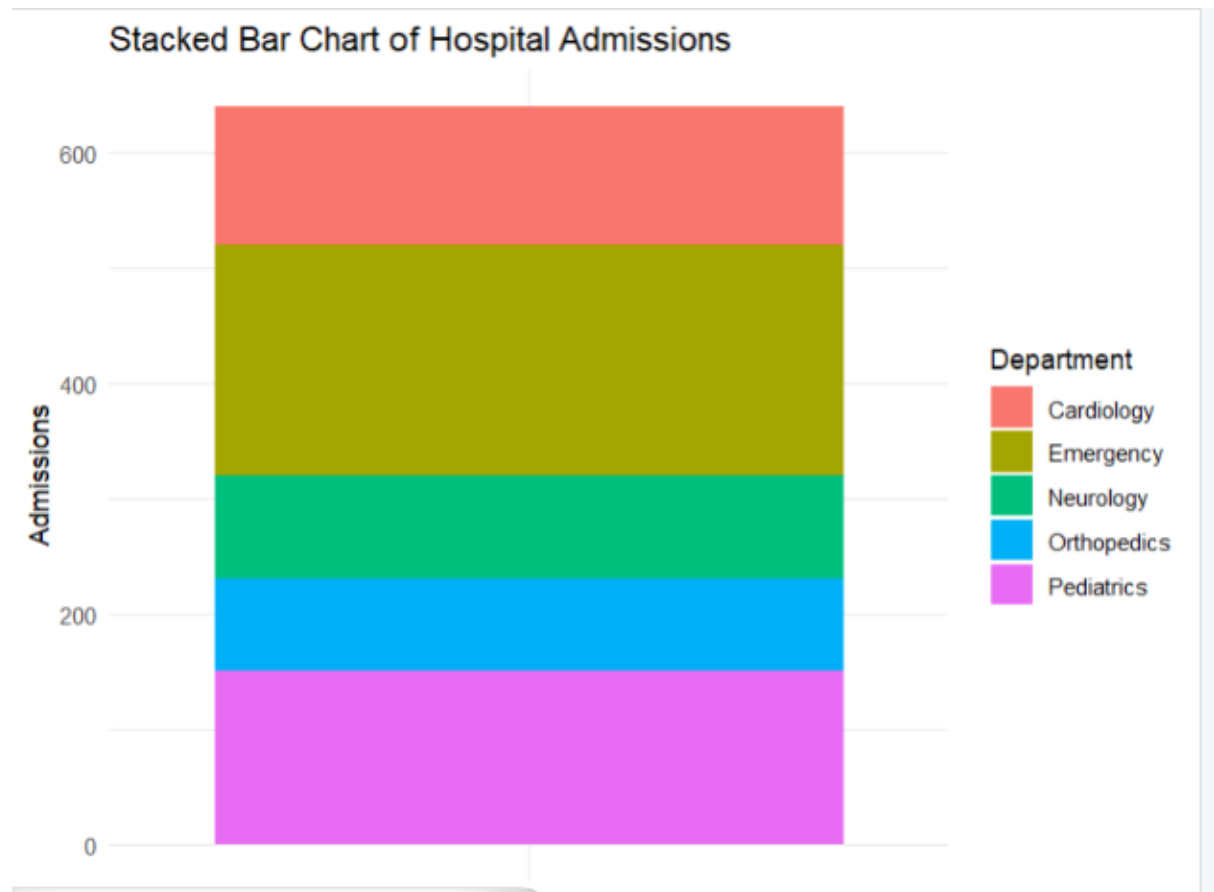
```
hospital$Year <- "2025"
```

```
ggplot(hospital,  
  aes(x = Year,  
    y = Admissions,  
    fill = Department)) +  
  geom_col() +  
  labs(
```

```

title = "Stacked Bar Chart of Hospital Admissions",
x = "Year",
y = "Admissions"
) +
theme_minimal()
output:

```



Tree map:

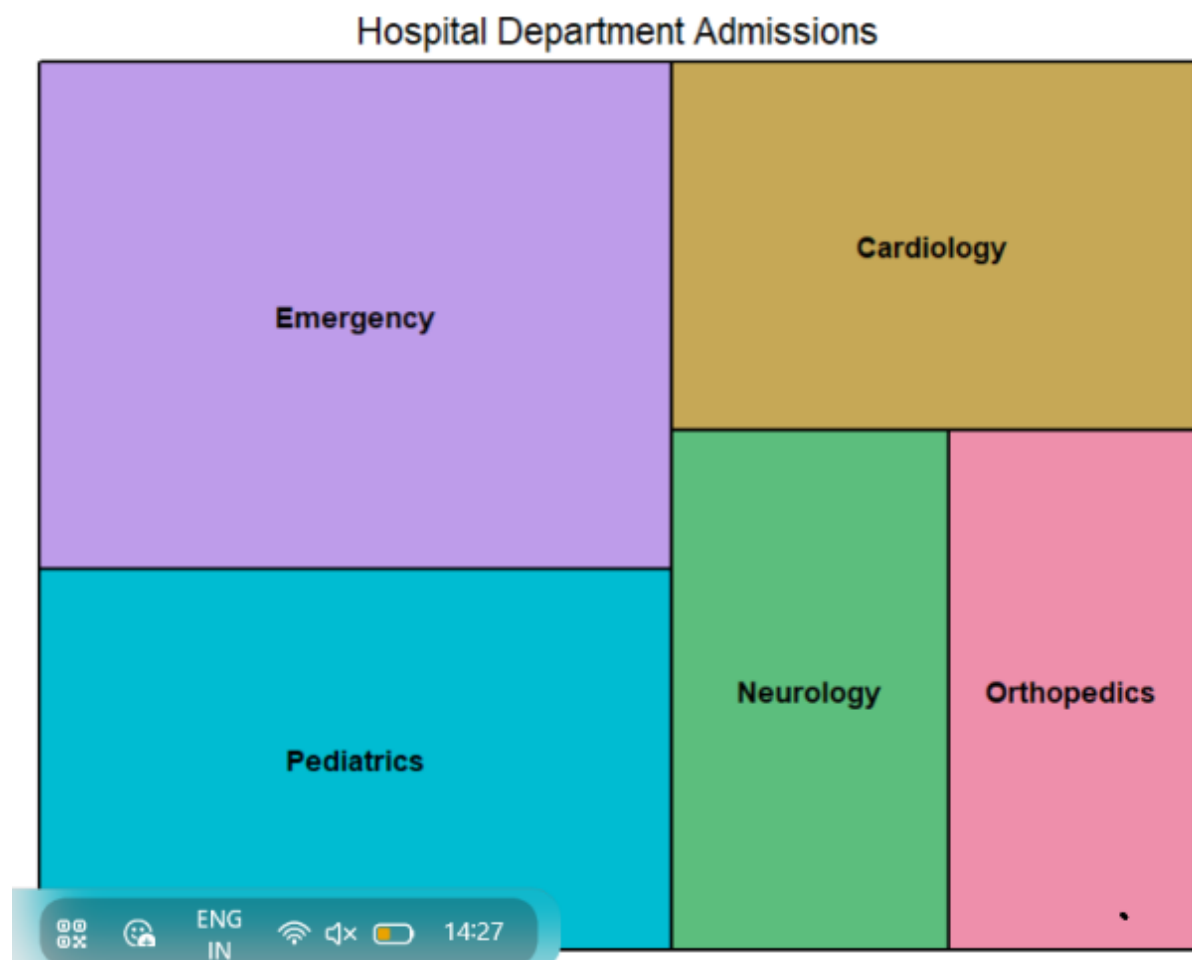
```
library(treemap)
```

```

treemap(
  hospital,
  index = "Department",
  vSize = "Admissions",
  title = "Hospital Department Admissions"
)

```

Output:

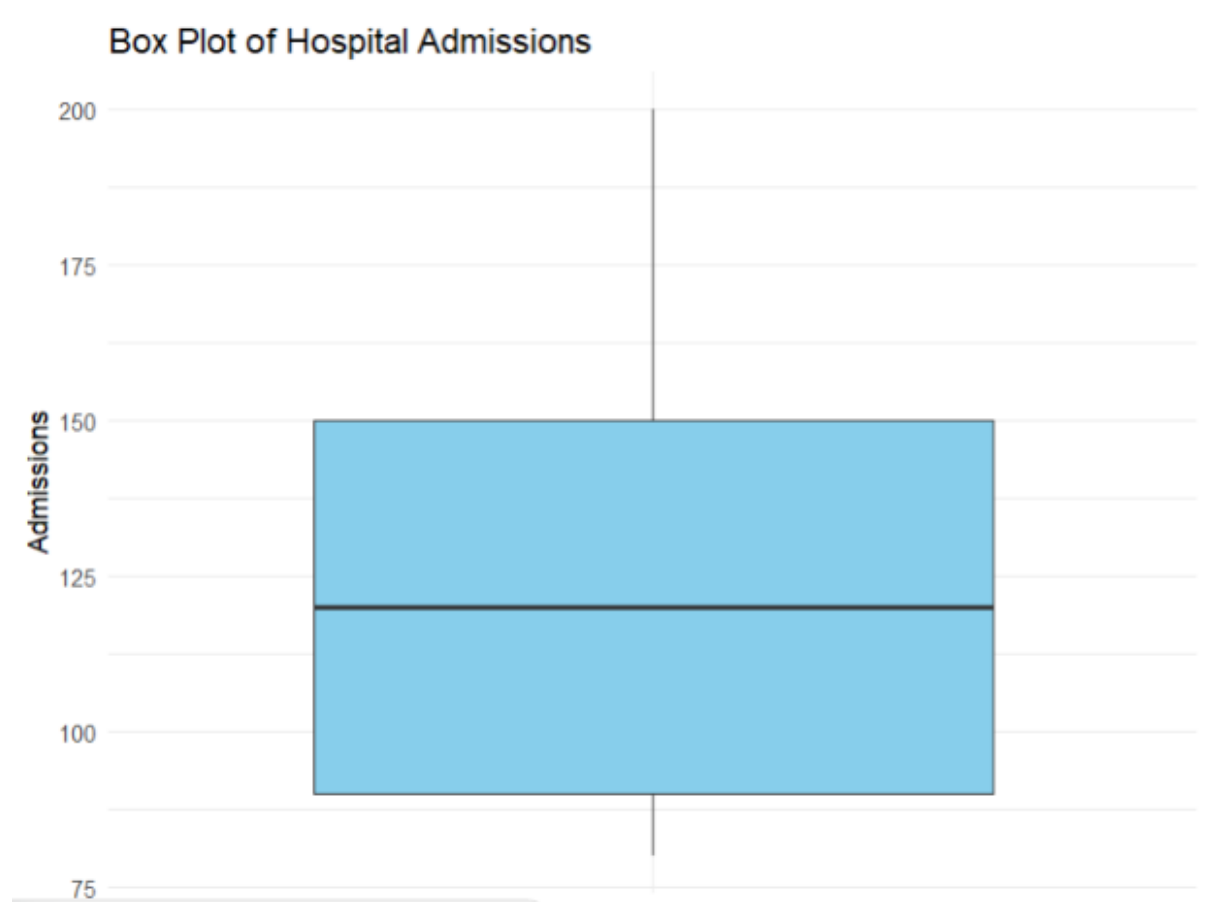


Histogram:

```
library(ggplot2)
```

```
ggplot(hospital,  
  aes(x = "",  
    y = Admissions)) +  
geom_boxplot(fill = "skyblue") +  
labs(  
  title = "Box Plot of Hospital Admissions",  
  x = "",  
  y = "Admissions"  
) +  
theme_minimal()
```

output:



Summary:

```
summary(hospital$Admissions)
```

output:

```
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
 80    90    120    128    150    200
```

2) E-Commerce Customer Spending Behaviour

R code:

```
# =====
```

```
# E-Commerce Customer Spending Behaviour
```

```
# =====
```

```
# Install packages if not already installed
```

```
# install.packages("ggplot2")
```

```
# install.packages("dplyr")
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
# -----
```

```
# Create Sample Dataset
```

```
# -----
```

```
set.seed(123)
```

```
ecommerce <- data.frame(
```

```
  CustomerID = 1:200,
```

```
  PurchaseAmount = round(rnorm(200, mean = 5000, sd = 1500),0),
```

```
  OrderFrequency = sample(1:20, 200, replace = TRUE),
```

```
  ProductCategory = sample(c("Electronics",
```

```
    "Fashion",
```

```
    "Home",
```

```
    "Sports",
```

```
    "Books"),
```

```
    200, replace = TRUE)
```

```
)
```

```
# Remove negative purchase amounts if any
```

```
ecommerce$PurchaseAmount[ecommerce$PurchaseAmount < 500] <- 500
```

```
# Display first records
```

```
head(ecommerce)
```

```
# =====
```

```
# 1. Histogram
```

```
# =====
```

```
ggplot(ecommerce,  
  aes(x = PurchaseAmount)) +  
  geom_histogram(binwidth = 500,  
    fill = "steelblue",  
    color = "black") +  
  labs(title = "Histogram of Purchase Amount",  
    x = "Purchase Amount",  
    y = "Number of Customers") +  
  theme_minimal()
```

```
# =====
```

```
# 2. Violin Plot
```

```
# =====
```

```
ggplot(ecommerce,  
  aes(x = ProductCategory,  
    y = PurchaseAmount,  
    fill = ProductCategory)) +  
  geom_violin(alpha = 0.7) +  
  labs(title = "Violin Plot of Purchase Amount by Category",  
    x = "Product Category",  
    y = "Purchase Amount") +  
  theme_minimal() +  
  theme(legend.position = "none")
```

```
# =====
```

```
# 3. Box Plot
```

```
# =====
```

```
ggplot(ecommerce,  
  aes(x = ProductCategory,  
    y = PurchaseAmount,  
    fill = ProductCategory)) +  
geom_boxplot() +  
labs(title = "Box Plot of Purchase Amount by Category",  
  x = "Product Category",  
  y = "Purchase Amount") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
# =====
```

```
# 4. Density Plot
```

```
# =====
```

```
ggplot(ecommerce,  
  aes(x = PurchaseAmount,  
    fill = ProductCategory)) +  
geom_density(alpha = 0.5) +  
labs(title = "Density Plot of Purchase Amount",  
  x = "Purchase Amount",  
  y = "Density") +  
theme_minimal()
```

```
# =====
```

```
# 5. Bar Chart
```

```
# =====
```



```
category_sales <- ecommerce %>%  
  group_by(ProductCategory) %>%  
  summarise(TotalSales = sum(PurchaseAmount))
```

```
ggplot(category_sales,  
  aes(x = ProductCategory,  
    y = TotalSales,  
    fill = ProductCategory)) +  
  geom_bar(stat = "identity") +  
  labs(title = "Total Sales by Product Category",  
    x = "Product Category",  
    y = "Total Sales") +  
  theme_minimal() +  
  theme(legend.position = "none")
```

```
# =====
```

```
# Summary Statistics
```

```
# =====
```

```
summary(ecommerce)
```

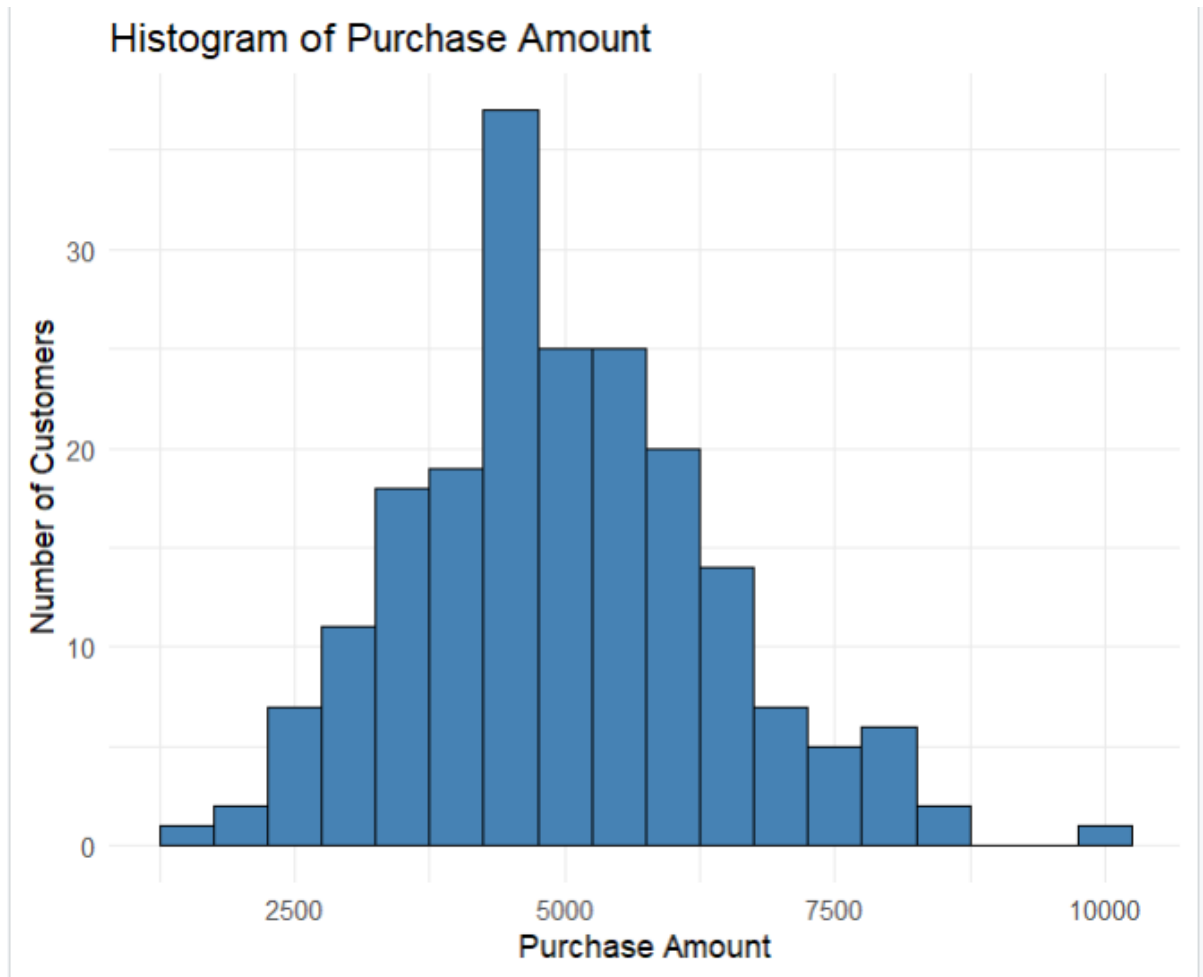
```
# Average spending by category
```

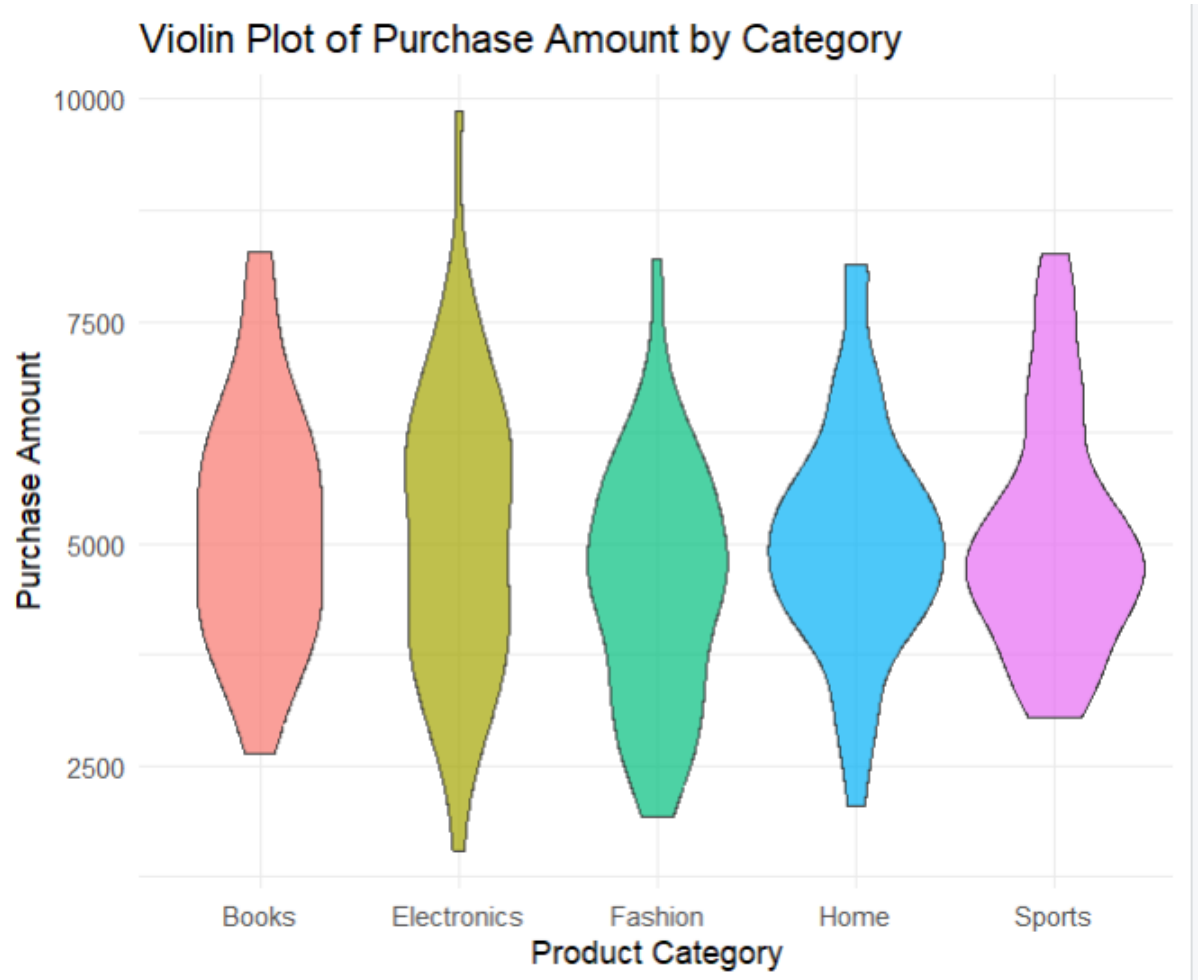
```
ecommerce %>%  
  group_by(ProductCategory) %>%  
  summarise(  
    AveragePurchase = mean(PurchaseAmount),  
    MaximumPurchase = max(PurchaseAmount),  
    MinimumPurchase = min(PurchaseAmount),
```

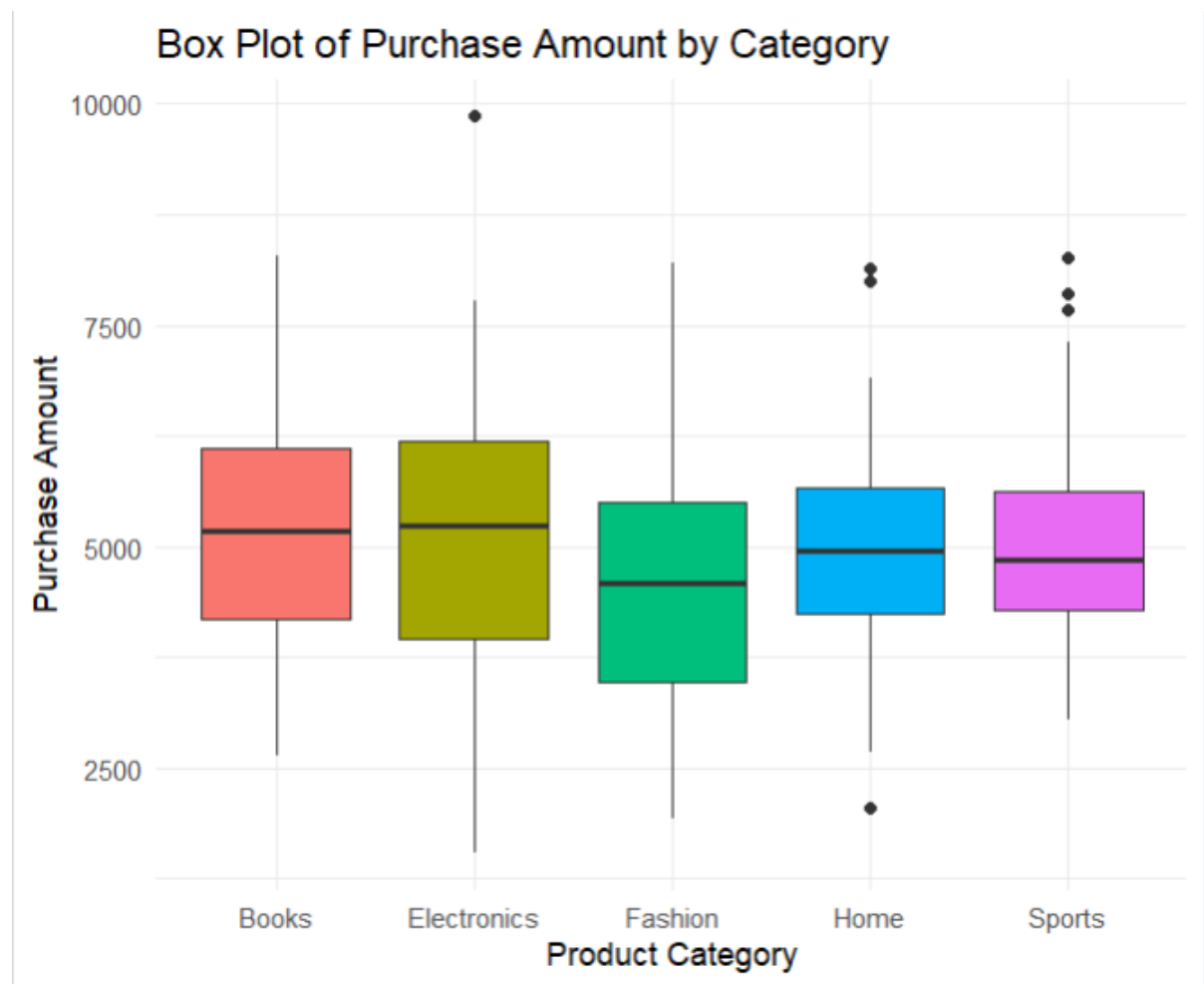
```
AverageOrders = mean(OrderFrequency)
```

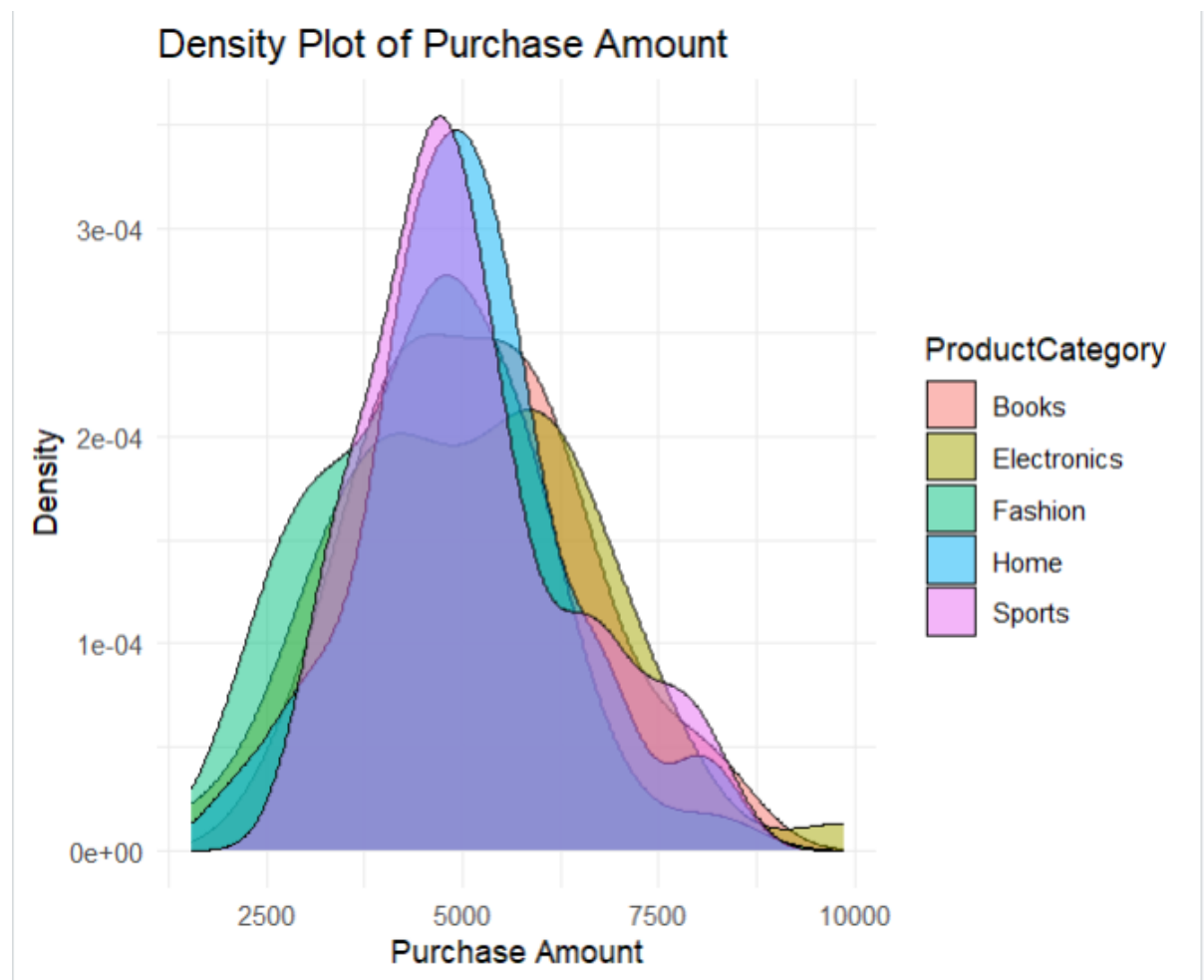
```
)
```

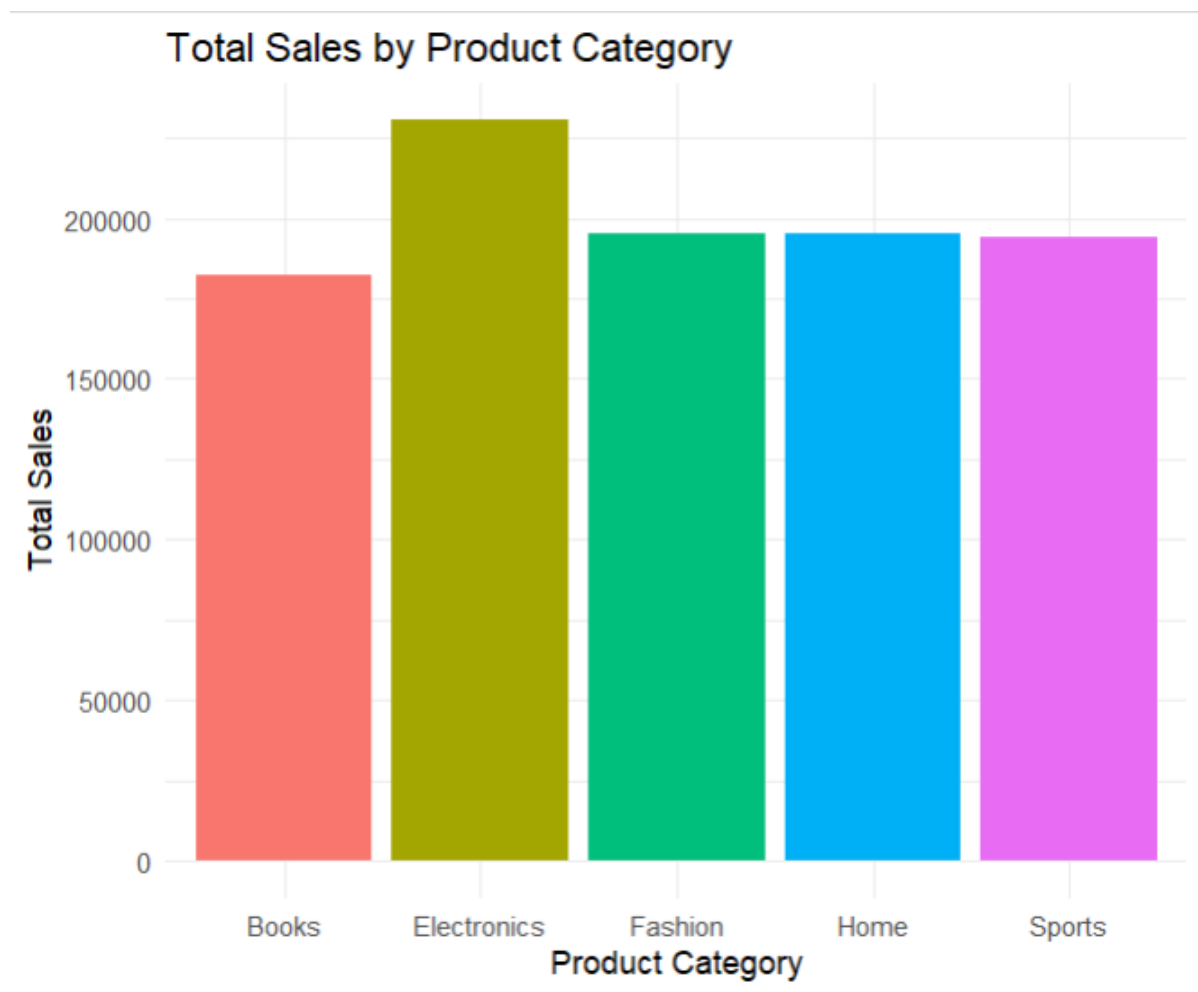
Output:











University Examination Performance Dashboard

A university wants to compare student marks across departments and identify score distributions, outliers, and top-performing classes. Generate different visualizations and evaluate which charts provide the clearest insights for academic decision-making.

Recommend improvements to the visualization dashboard.

R code:

```
# =====  
# University Examination Performance Dashboard  
# =====  
  
# Install packages if not already installed  
# install.packages("ggplot2")
```

```

# install.packages("dplyr")

library(ggplot2)

library(dplyr)

# -----

# Create Sample Dataset

# -----

set.seed(123)

students <- data.frame(
  StudentID = 1:300,
  Department = sample(c("Computer Science",
                        "Mechanical",
                        "Electrical",
                        "Civil",
                        "Electronics"),
                      300, replace = TRUE),
  Marks = round(rnorm(300, mean = 70, sd = 12),0),
  Semester = sample(c("Semester 1",
                      "Semester 2",
                      "Semester 3"),
                    300, replace = TRUE)
)

# Limit marks between 0 and 100
students$Marks[students$Marks < 0] <- 0
students$Marks[students$Marks > 100] <- 100

```

```

# Display first records
head(students)

# =====

# 1. Histogram
# =====

ggplot(students,
       aes(x = Marks)) +
  geom_histogram(binwidth = 5,
                 fill = "steelblue",
                 color = "black") +
  labs(title = "Distribution of Student Marks",
       x = "Marks",
       y = "Number of Students") +
  theme_minimal()

# =====

# 2. Box Plot
# =====

ggplot(students,
       aes(x = Department,
          y = Marks,
          fill = Department)) +
  geom_boxplot() +
  labs(title = "Marks Distribution by Department",
       x = "Department",

```



```
y = "Marks") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
# =====
```

```
# 3. Violin Plot
```

```
# =====
```

```
ggplot(students,  
  aes(x = Department,  
    y = Marks,  
    fill = Department)) +  
geom_violin(alpha = 0.7) +  
labs(title = "Violin Plot of Marks by Department",  
  x = "Department",  
  y = "Marks") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
# =====
```

```
# 4. Bar Chart
```

```
# Average Marks by Department
```

```
# =====
```

```
avg_marks <- students %>%  
  group_by(Department) %>%  
  summarise(AverageMarks = mean(Marks))
```

```
ggplot(avg_marks,
```

```

aes(x = Department,
    y = AverageMarks,
    fill = Department)) +
geom_bar(stat = "identity") +
labs(title = "Average Marks by Department",
     x = "Department",
     y = "Average Marks") +
theme_minimal() +
theme(legend.position = "none")

```

```
# =====
```

```
# 5. Scatter Plot
```

```
# Marks vs Student ID
```

```
# =====
```

```

ggplot(students,
       aes(x = StudentID,
           y = Marks,
           color = Department)) +
geom_point(size = 2) +
labs(title = "Student Marks Distribution",
     x = "Student ID",
     y = "Marks") +
theme_minimal()

```

```
# =====
```

```
# Summary Statistics
```

```
# =====
```

```
summary(students)
```

```
# Department-wise Performance
```

```
students %>%
```

```
  group_by(Department) %>%
```

```
  summarise(
```

```
    AverageMarks = mean(Marks),
```

```
    HighestMarks = max(Marks),
```

```
    LowestMarks = min(Marks),
```

```
    TotalStudents = n()
```

```
  )
```

```
# Identify Top Performing Department
```

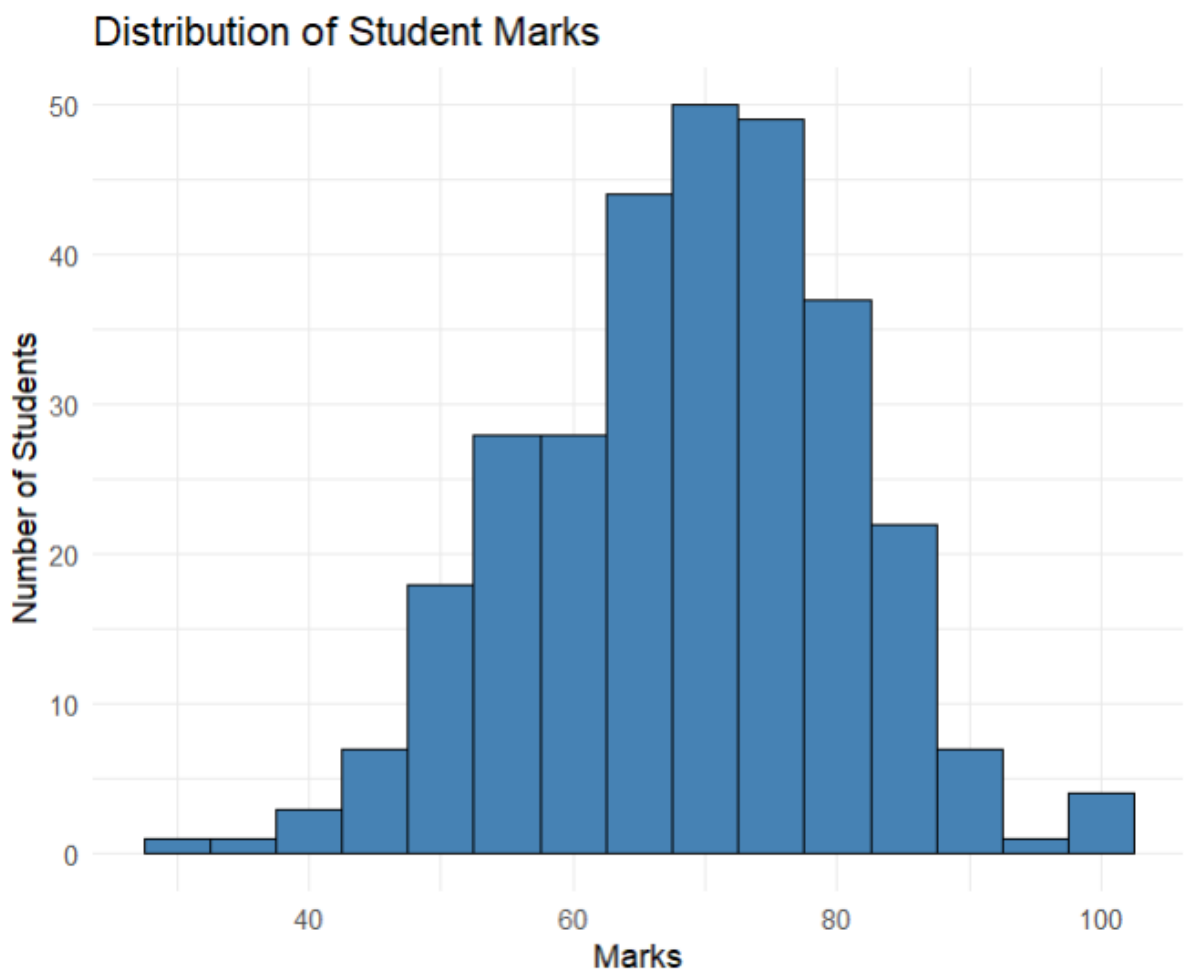
```
students %>%
```

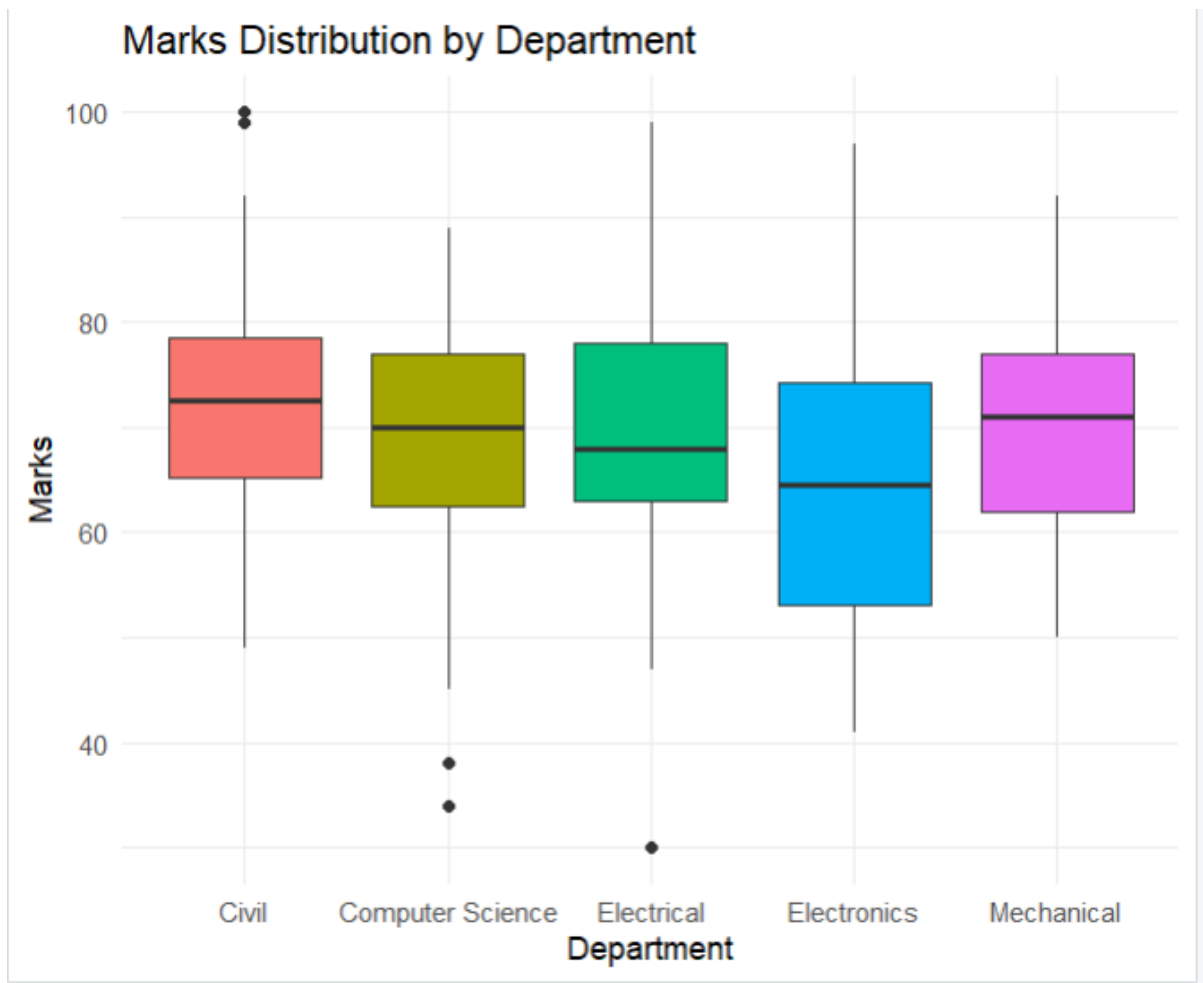
```
  group_by(Department) %>%
```

```
  summarise(AverageMarks = mean(Marks)) %>%
```

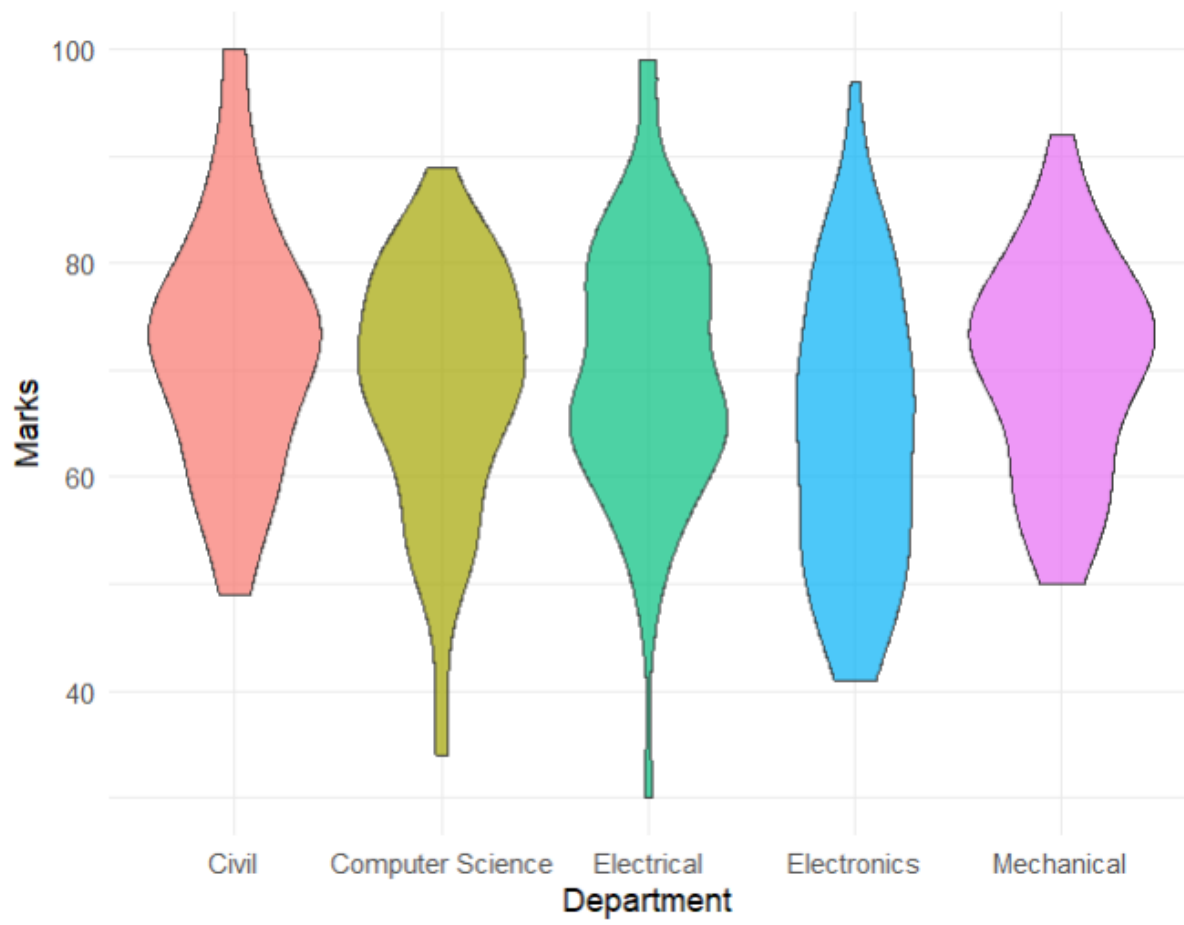
```
  arrange(desc(AverageMarks))
```

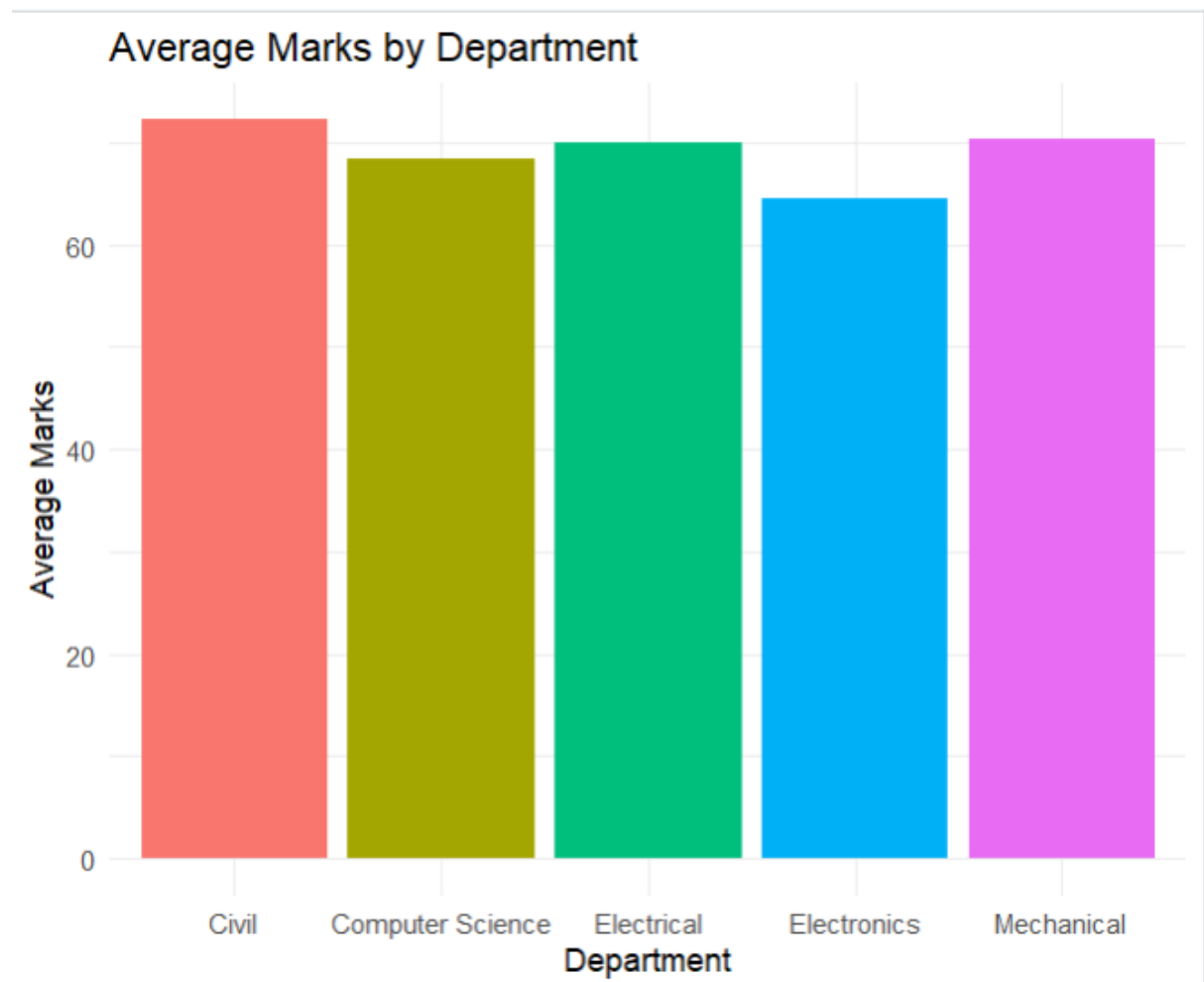
```
output:
```

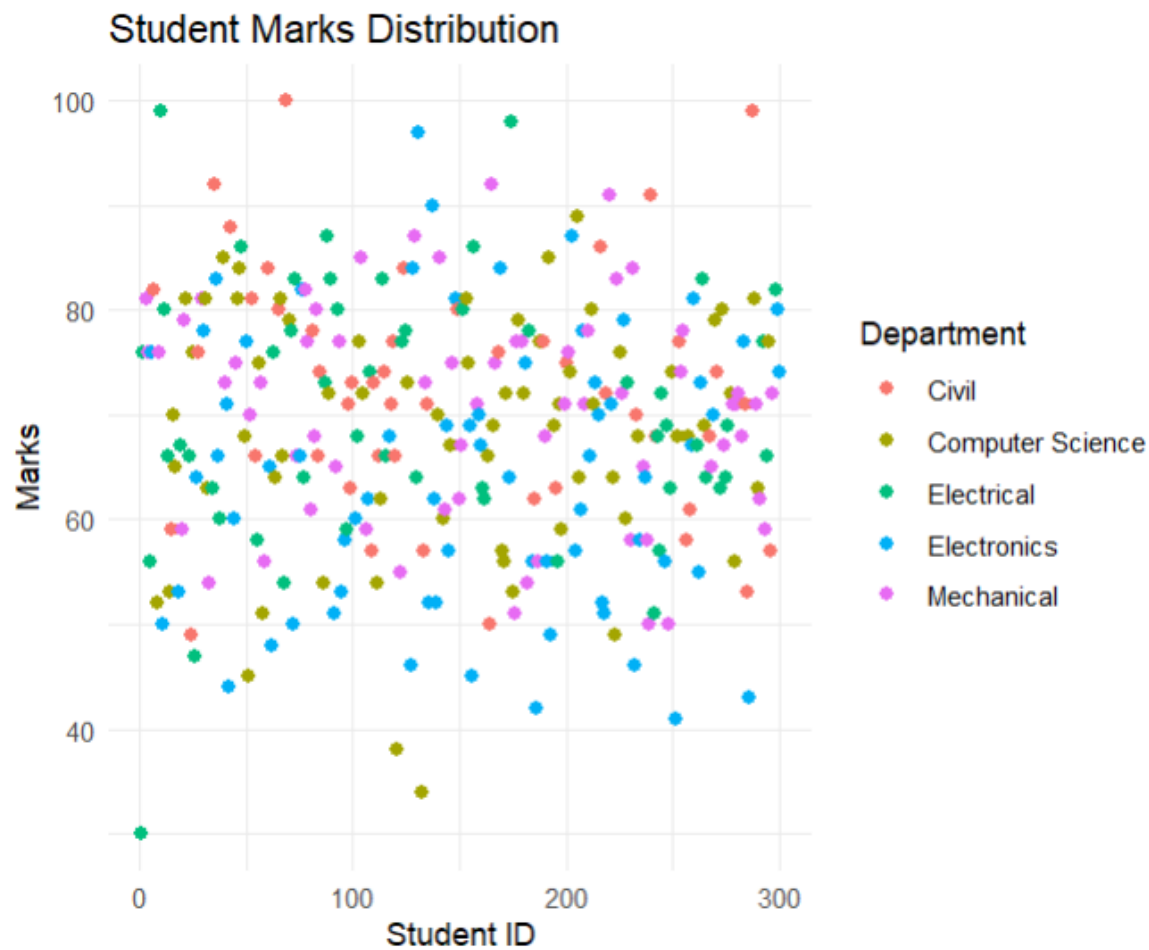




Violin Plot of Marks by Department







Climate and Rainfall Distribution Study

Meteorological data contains monthly rainfall, temperature, and humidity for multiple cities. Produce histograms, density plots, box plots, and bar charts. Evaluate which visualization best represents seasonal rainfall variation and climate distribution. Compare at least three visualizations and justify the most suitable one for environmental researchers.

R code:

```
# =====
# Climate and Rainfall Distribution Study
# =====

# Install packages if not already installed
```



```
# install.packages("ggplot2")
# install.packages("dplyr")

library(ggplot2)
library(dplyr)

# -----
# Create Sample Dataset
# -----

set.seed(123)

climate <- data.frame(
  City = sample(c("Hyderabad",
                  "Chennai",
                  "Mumbai",
                  "Delhi",
                  "Bangalore"),
                240, replace = TRUE),

  Month = sample(month.abb, 240, replace = TRUE),

  Rainfall = round(rnorm(240, mean = 120, sd = 40),1),

  Temperature = round(rnorm(240, mean = 30, sd = 5),1),

  Humidity = round(rnorm(240, mean = 70, sd = 10),1)
)
```

```
# Remove negative rainfall values
climate$Rainfall[climate$Rainfall < 0] <- 0
```

```
# Display first records
head(climate)
```

```
# =====
```

```
# 1. Histogram
```

```
# Rainfall Distribution
```

```
# =====
```

```
ggplot(climate,
       aes(x = Rainfall)) +
  geom_histogram(binwidth = 10,
                 fill = "steelblue",
                 color = "black") +
  labs(title = "Histogram of Monthly Rainfall",
       x = "Rainfall (mm)",
       y = "Frequency") +
  theme_minimal()
```

```
# =====
```

```
# 2. Density Plot
```

```
# =====
```

```
ggplot(climate,
       aes(x = Rainfall,
           fill = City)) +
  geom_density(alpha = 0.5) +
```

```
labs(title = "Density Plot of Rainfall by City",
      x = "Rainfall (mm)",
      y = "Density") +
theme_minimal()
```

```
# =====
```

```
# 3. Box Plot
```

```
# =====
```

```
ggplot(climate,
       aes(x = City,
           y = Rainfall,
           fill = City)) +
geom_boxplot() +
labs(title = "Rainfall Distribution by City",
      x = "City",
      y = "Rainfall (mm)") +
theme_minimal() +
theme(legend.position = "none")
```

```
# =====
```

```
# 4. Bar Chart
```

```
# Average Rainfall by City
```

```
# =====
```

```
avg_rainfall <- climate %>%
  group_by(City) %>%
  summarise(AverageRainfall = mean(Rainfall))
```

```

ggplot(avg_rainfall,
  aes(x = City,
    y = AverageRainfall,
    fill = City)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Rainfall by City",
    x = "City",
    y = "Average Rainfall (mm)") +
  theme_minimal() +
  theme(legend.position = "none")

# =====

# Summary Statistics

# =====

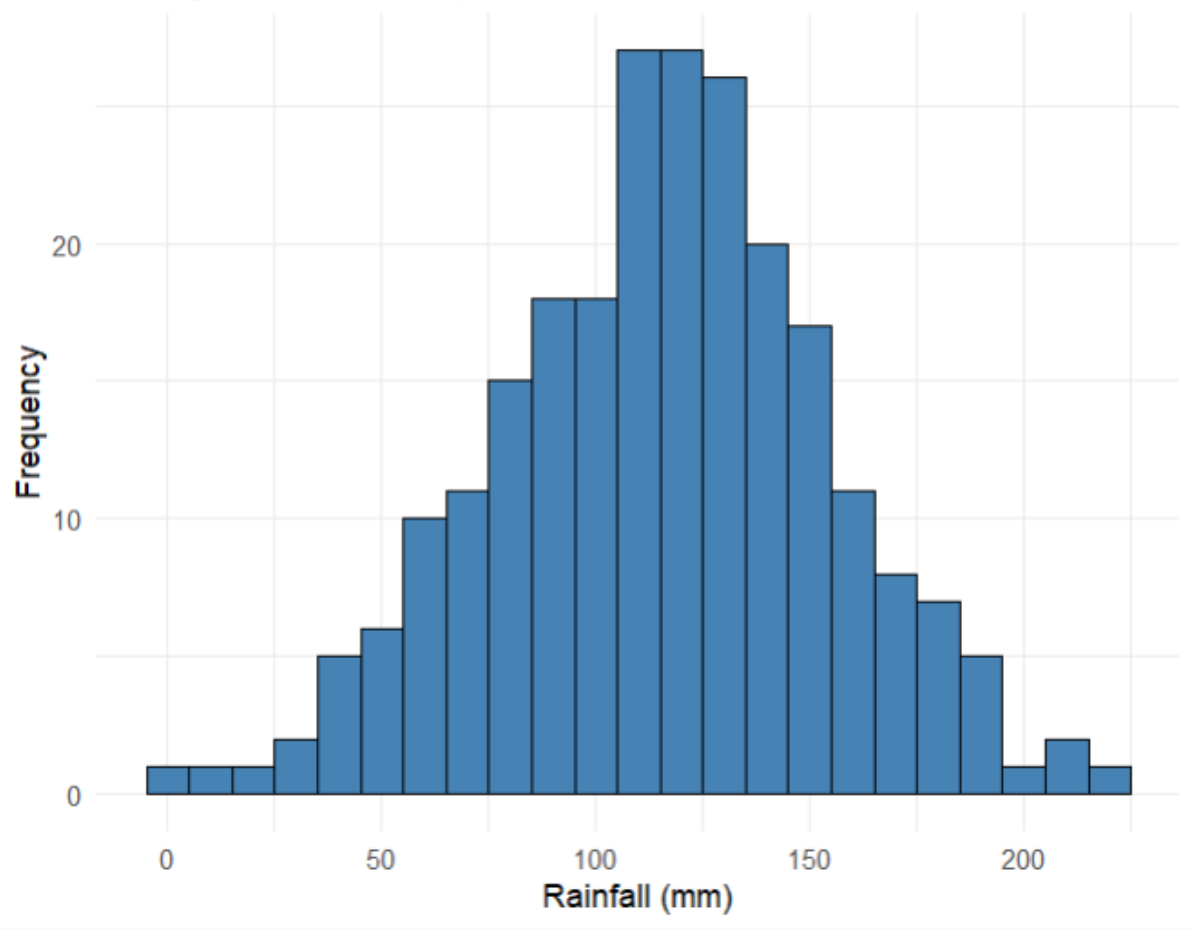
summary(climate)

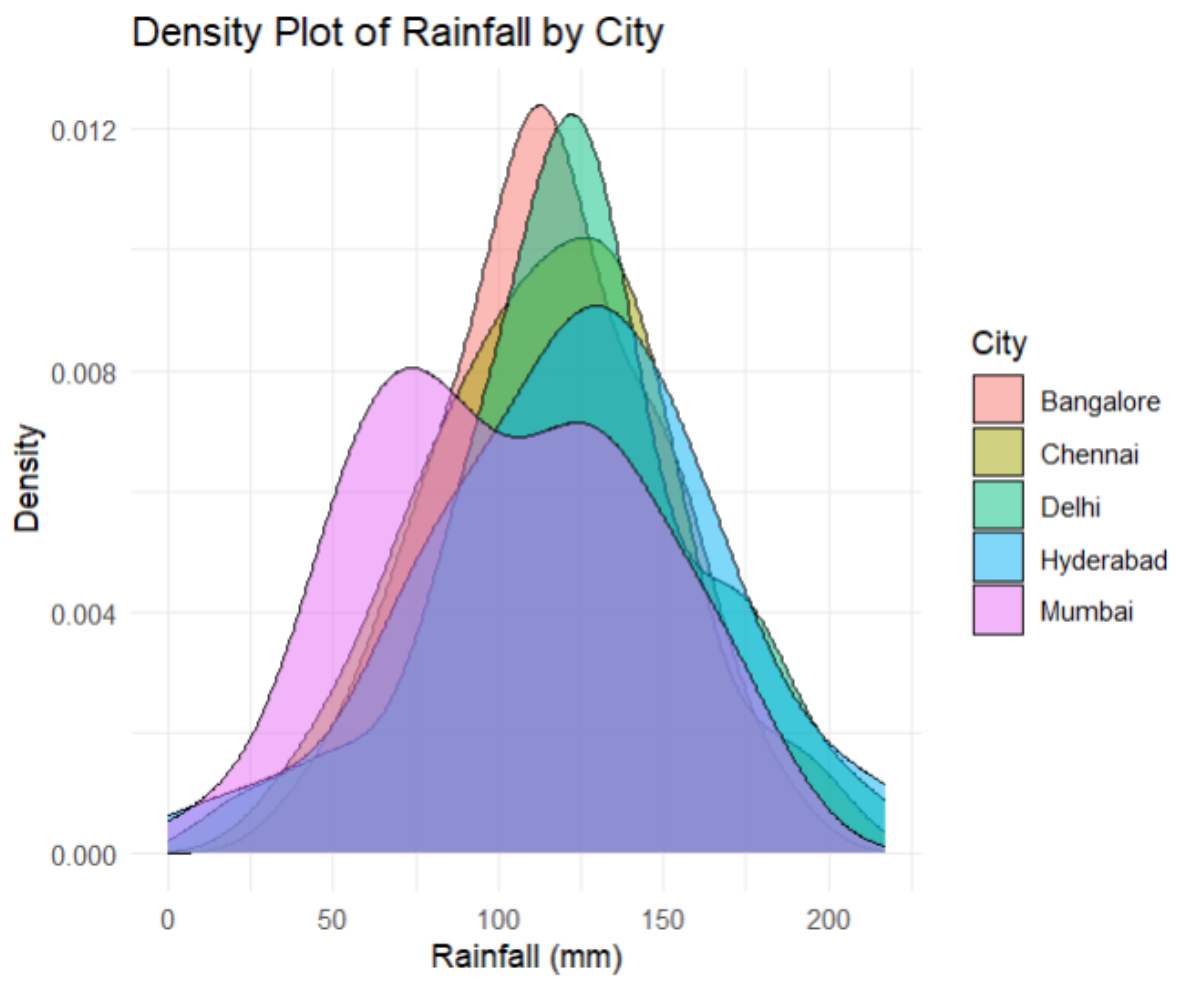
# City-wise Climate Summary
climate %>%
  group_by(City) %>%
  summarise(
    AverageRainfall = mean(Rainfall),
    AverageTemperature = mean(Temperature),
    AverageHumidity = mean(Humidity),
    MaximumRainfall = max(Rainfall),
    MinimumRainfall = min(Rainfall)
  )

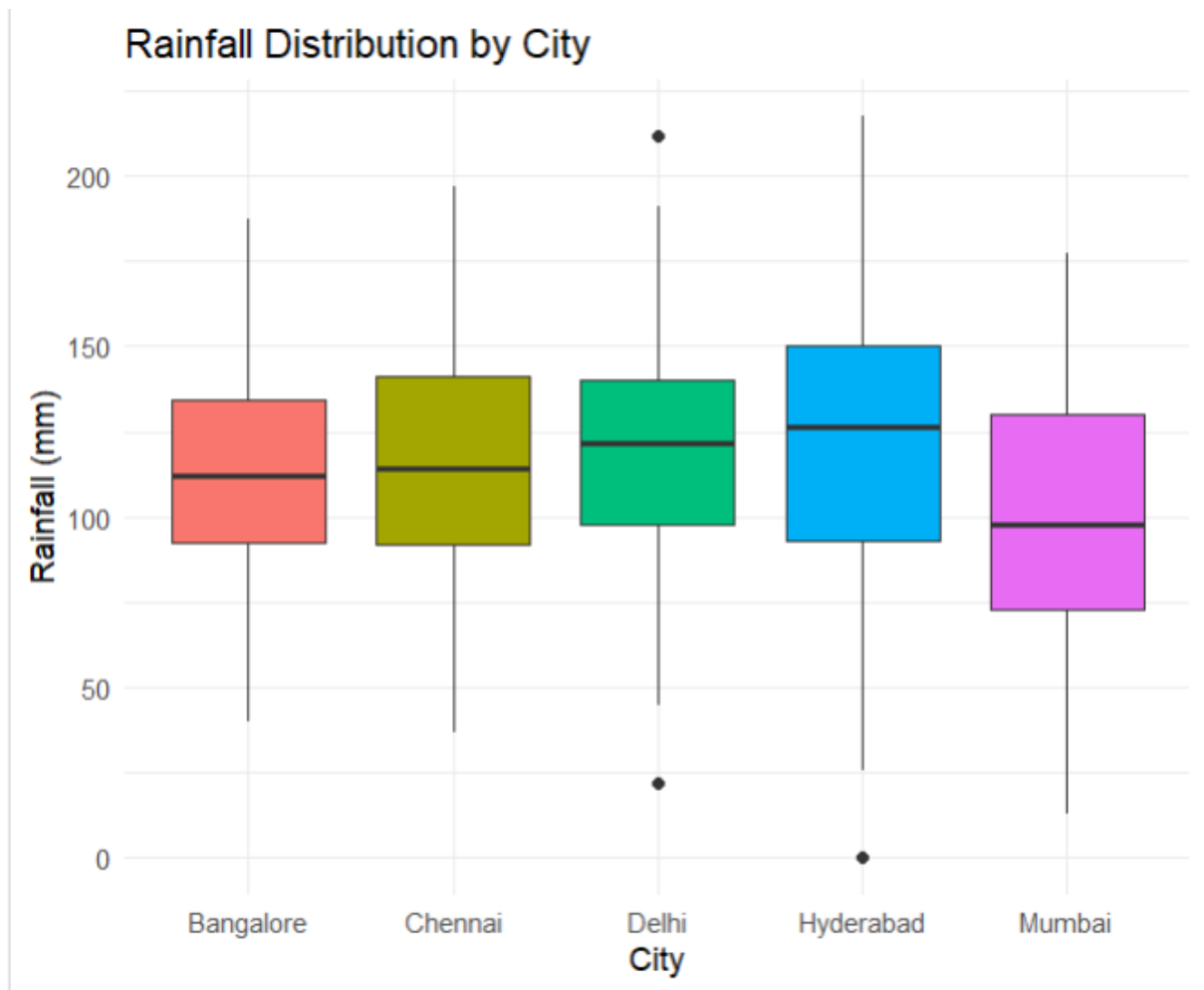
```

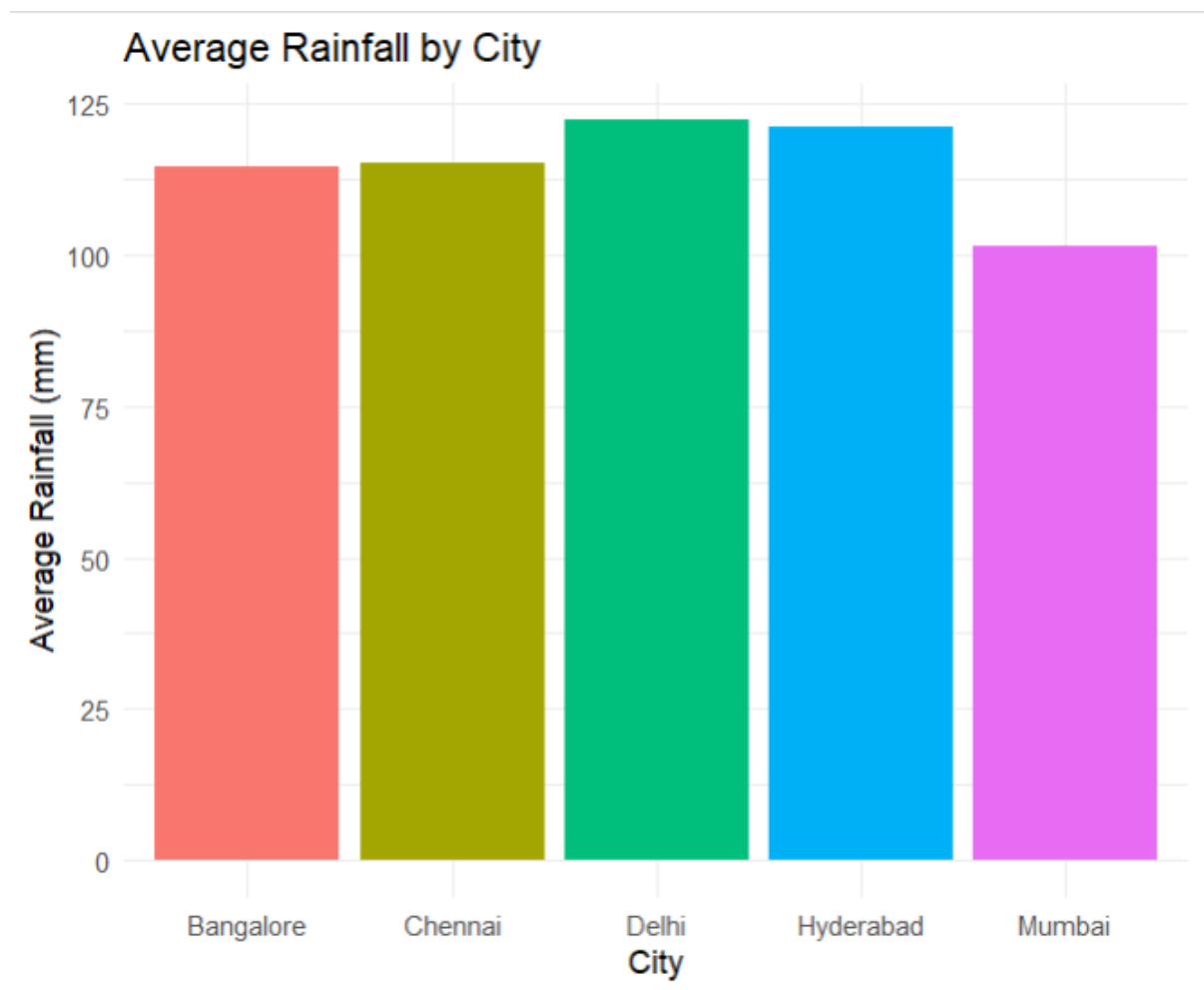
Output:

Histogram of Monthly Rainfall









Bank Loan Risk Assessment

A financial institution has customer loan amounts, repayment history, income levels, and credit scores. Develop visualizations showing loan distributions and customer segments. Evaluate which visualization best supports loan approval decisions by highlighting risk patterns, skewness, and outliers. Recommend the visualization that should be included in a management dashboard.

R code:

```
# =====  
# Bank Loan Risk Assessment  
# =====  
  
# Install packages if not already installed
```


)

```
# Remove unrealistic values

loan_data$LoanAmount[loan_data$LoanAmount < 50000] <- 50000

loan_data$Income[loan_data$Income < 10000] <- 10000

loan_data$CreditScore[loan_data$CreditScore < 300] <- 300

loan_data$CreditScore[loan_data$CreditScore > 900] <- 900
```

```
# Display first records
```

```
head(loan_data)
```

```
# =====
```

```
# 1. Histogram
```

```
# Loan Amount Distribution
```

```
# =====
```

```
ggplot(loan_data,
       aes(x = LoanAmount)) +
  geom_histogram(binwidth = 50000,
                 fill = "steelblue",
                 color = "black") +
  labs(title = "Distribution of Loan Amount",
       x = "Loan Amount",
       y = "Number of Customers") +
  theme_minimal()
```

```
# =====
```

```
# 2. Density Plot
```

```
# Credit Score Distribution
```

```
# =====
```

```
ggplot(loan_data,  
  aes(x = CreditScore,  
    fill = RepaymentStatus)) +  
geom_density(alpha = 0.5) +  
labs(title = "Credit Score Distribution by Repayment Status",  
  x = "Credit Score",  
  y = "Density") +  
theme_minimal()
```

```
# =====
```

```
# 3. Box Plot
```

```
# Loan Amount by Repayment Status
```

```
# =====
```

```
ggplot(loan_data,  
  aes(x = RepaymentStatus,  
    y = LoanAmount,  
    fill = RepaymentStatus)) +  
geom_boxplot() +  
labs(title = "Loan Amount by Repayment Status",  
  x = "Repayment Status",  
  y = "Loan Amount") +  
theme_minimal() +  
theme(legend.position = "none")
```

```
# =====
```

```
# 4. Bar Chart
```

```
# Customer Segments
```

```
# =====
```

```

segment_count <- loan_data %>%
  group_by(RepaymentStatus) %>%
  summarise(Customers = n())

ggplot(segment_count,
  aes(x = RepaymentStatus,
    y = Customers,
    fill = RepaymentStatus)) +
  geom_bar(stat = "identity") +
  labs(title = "Customer Segments by Repayment Status",
    x = "Repayment Status",
    y = "Number of Customers") +
  theme_minimal() +
  theme(legend.position = "none")

```

```
# =====
```

```
# 5. Scatter Plot
```

```
# Income vs Loan Amount
```

```
# =====
```

```

ggplot(loan_data,
  aes(x = Income,
    y = LoanAmount,
    color = RepaymentStatus)) +
  geom_point(size = 2) +
  labs(title = "Income vs Loan Amount",
    x = "Income",
    y = "Loan Amount") +

```

```
theme_minimal()
```

```
# =====
```

```
# Summary Statistics
```

```
# =====
```

```
summary(loan_data)
```

```
# Repayment Status Summary
```

```
loan_data %>%
```

```
  group_by(RepaymentStatus) %>%
```

```
  summarise(
```

```
    AverageLoan = mean(LoanAmount),
```

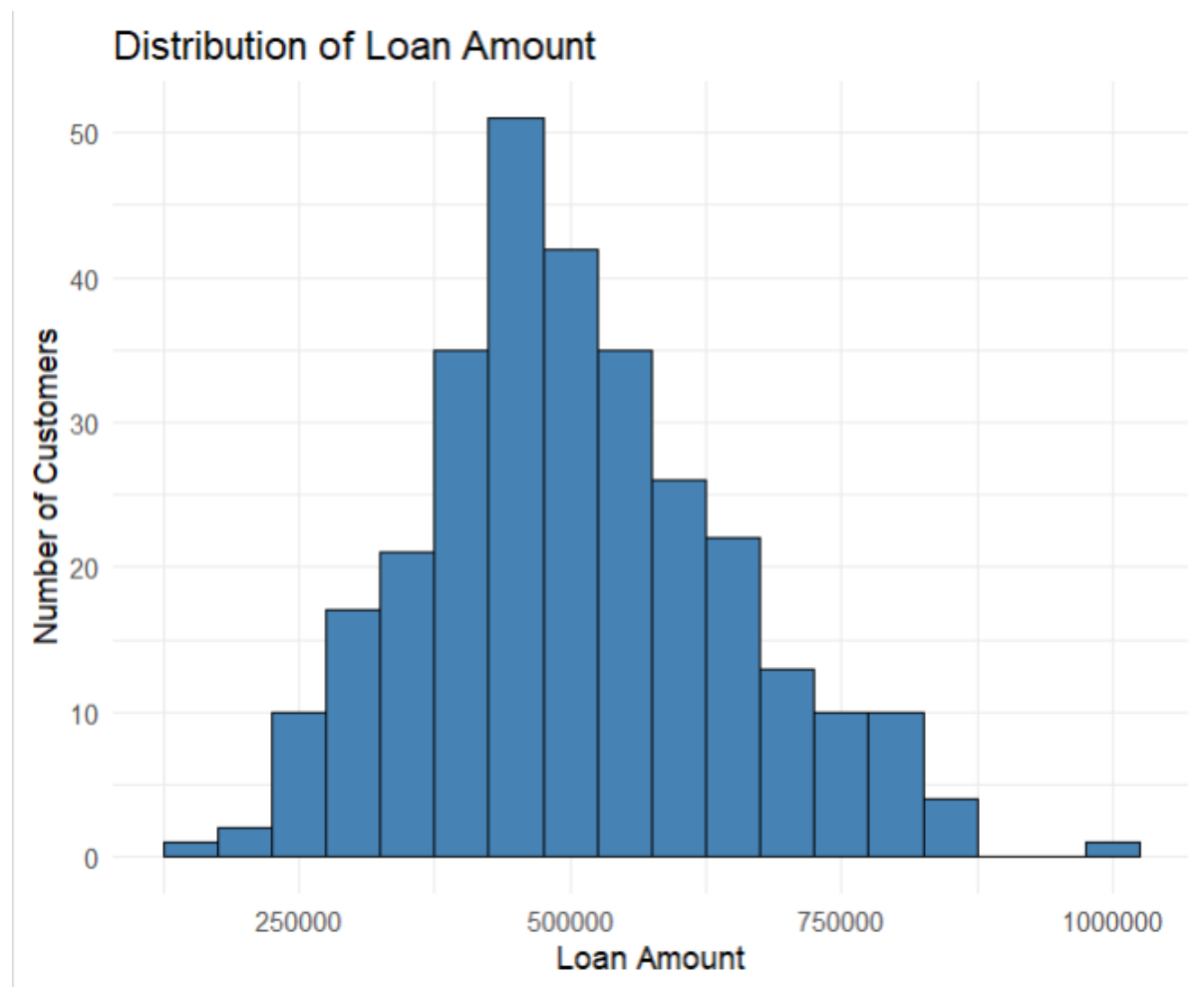
```
    AverageIncome = mean(Income),
```

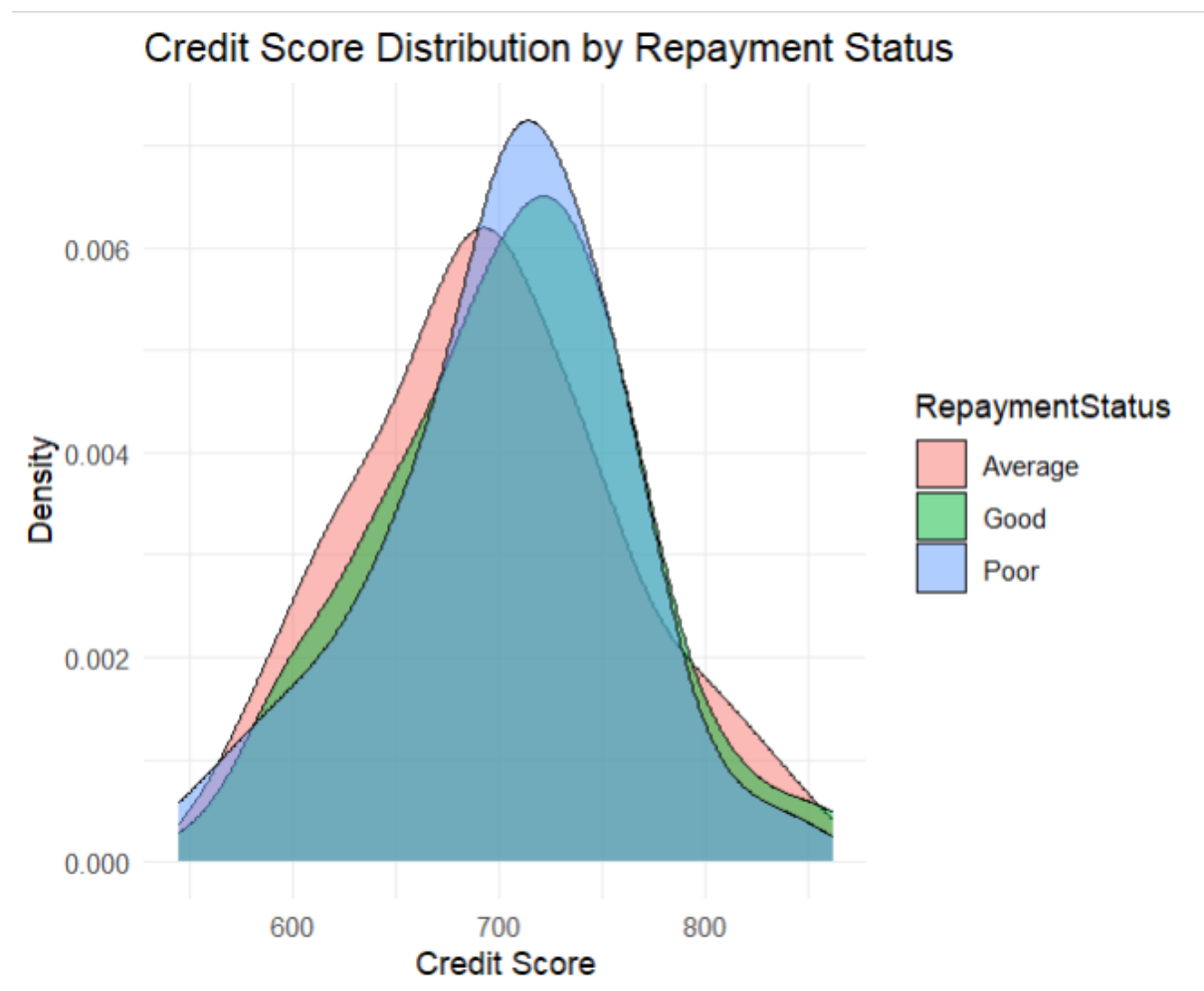
```
    AverageCreditScore = mean(CreditScore),
```

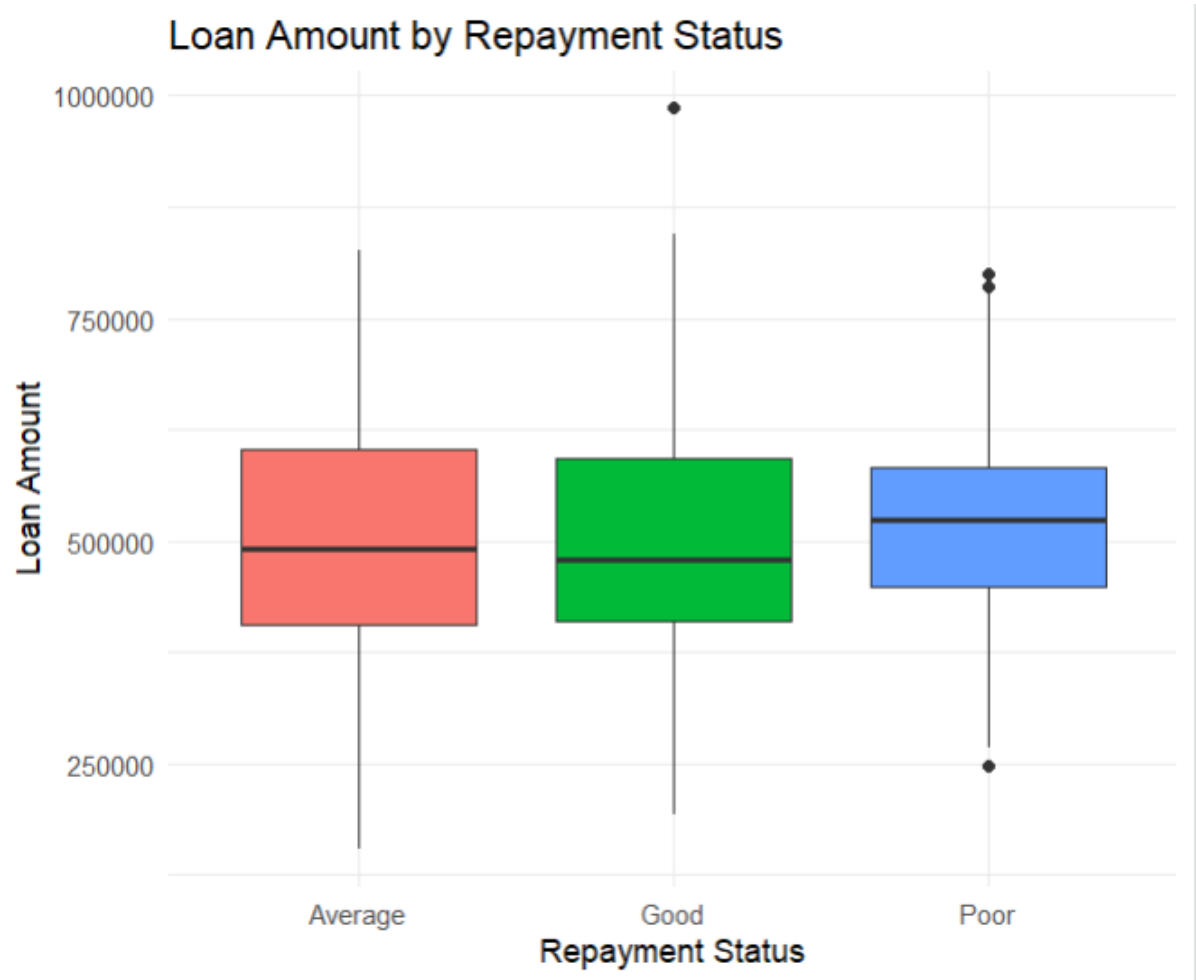
```
    CustomerCount = n()
```

```
)
```

Output:







Customer Segments by Repayment Status

